PARADATA FORMAT GUIDE

SurveyCTO and ODK Audit Log Structures, Field Conventions, and Parsing Strategies

Project: Automated Extraction and Analysis of Timing-Based Paradata Features for Survey Quality Assurance

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1. Purpose and Scope

This document provides a comprehensive reference for the audit log formats generated by SurveyCTO and Open Data Kit (ODK) — the two digital data collection platforms supported by the paradata feature extraction pipeline developed in this project. It documents the exact column structures, timestamp formats, event taxonomies, and field naming conventions used by each platform, and defines the Common Intermediate Representation (CIR) into which both formats are normalized before feature extraction.

This guide serves two audiences: (1) developers extending the pipeline to support additional platforms, who need to understand the CIR schema and transformation logic; and (2) survey researchers seeking to work directly with raw paradata, who need a clear reference for what each log column contains and how to interpret it.

Scope

This guide covers SurveyCTO audit logs (CSV format, event-based, millisecond timestamps) and ODK audit logs (CSV format, question-based, ISO 8601 timestamps). Google Forms and other platforms are

currently out of scope. The minimum log requirements for pipeline compatibility are documented in Section 5.

2. SurveyCTO Audit Log Format

SurveyCTO automatically generates an audit log CSV file for each completed interview when audit logging is enabled in the form definition. The audit log captures every form event at millisecond precision, tied to the XPath identifier of the field involved.

2.1 File Structure

The audit log is a plain CSV file with a header row followed by one row per event. The file is named after the submission UUID and is attached to the submission record. Each row represents a single discrete event in the interview session.

2.2 Column Definitions

Column Name	Data Type	Description
`event`	String	The type of form event recorded. See Section 2.3 for the complete event taxonomy.
`node`	String	The XPath identifier of the field associated with this event (e.g., /data/household/q_age). Empty for form-level events such as form-start and form-end.
`start`	Integer (ms)	Unix epoch timestamp in milliseconds marking the beginning of the event.
`end`	Integer (ms)	Unix epoch timestamp in milliseconds marking the end of the event. May be null for instantaneous events.
`latitude`	Float	GPS latitude recorded at the time of the event, if location capture is enabled. Not used by this pipeline.
`longitude`	Float	GPS longitude. Not used by this pipeline.
`accuracy`	Float	GPS accuracy in metres. Not used by this pipeline.
`altitude`	Float	GPS altitude in metres. Not used by this pipeline.

Table 1. SurveyCTO audit log column definitions.

2.3 Event Taxonomy

SurveyCTO records the following event types. Understanding this taxonomy is essential for correct timestamp extraction and duration computation.

Event Type	Meaning
`form-start`	The interview session began. Marks the absolute start of the interview. The node field is empty.
`form-resume`	The interview was resumed after being paused or saved. Indicates an interrupted session.
`question`	The enumerator navigated to a field. start is when the field became active; end is when they left it.
`constraint`	A validation constraint error was triggered on the current field. Indicates incorrect or out-of-range input.
`save`	The form was saved (but not submitted). May occur mid-interview if connectivity is lost.
`end`	The interview session ended. Marks the absolute end of the interview. The node field is empty.
`location`	A GPS location capture event. Not relevant to timing feature extraction.

Table 2. SurveyCTO event taxonomy.

2.4 Example Log Rows

The following illustrates a realistic SurveyCTO audit log excerpt for a short interview:

```
event,node,start,end
form-start,,1711440000000,1711440001200
question,/data/consent,1711440001200,1711440009800
question,/data/household/q_age,1711440009800,1711440017400
constraint,/data/household/q_age,1711440014200,1711440014200
question,/data/household/q_age,1711440014200,1711440019600
question,/data/income/q_monthly,1711440019600,1711440038200
end,,1711440038200,1711440039000
```

2.5 Known Quirks and Edge Cases

- Some SurveyCTO versions record duplicate event rows for the same field when the enumerator navigates back. The parser deduplicates by keeping the last event for each field visit.
- The end timestamp of one event and the start timestamp of the next event are typically identical — this is by design and does not represent a gap.

- If the form is saved and resumed, a form-resume event appears between question events. The pipeline treats the gap between save and form-resume as a session pause, not question-level idle time.
- The node field uses XPath notation (/data/group/field). The parser extracts the final path component as the field_name in the CIR.
- GPS columns are always present in the CSV header even when location capture is disabled. They will be null for all rows in such cases.

3. ODK Audit Log Format

Open Data Kit (ODK) Collect generates an audit log CSV when audit logging is configured in the XLSForm. ODK's event taxonomy and timestamp format differ meaningfully from SurveyCTO, requiring distinct parsing logic before the data can be normalized into the CIR.

3.1 File Structure

The ODK audit log is a CSV file stored within the submission package. Unlike SurveyCTO, ODK uses a question-centric event model rather than a granular event-type model — each row typically represents a question view rather than a discrete form action.

3.2 Column Definitions

Column Name	Data Type	Description
`event`	String	The type of event. ODK uses a different taxonomy from SurveyCTO. See Section 3.3.
`node`	String	The XPath field identifier. Same format as SurveyCTO (/data/group/field).
`start`	String (ISO 8601)	Timestamp in ISO 8601 format (YYYY-MM-DDTHH:MM:SS.sss). Must be converted to epoch milliseconds by the parser.
`end`	String (ISO 8601)	End timestamp in ISO 8601 format. May be absent for instantaneous events.
`latitude`	Float	GPS latitude if location capture enabled. Not used by pipeline.
`longitude`	Float	GPS longitude. Not used by pipeline.
`accuracy`	Float	GPS accuracy in metres. Not used by pipeline.

old-value	String	Previous value of a field before edit. Only present in some ODK versions.
new-value	String	New value after edit. Only present in some ODK versions.

Table 3. ODK audit log column definitions.

3.3 ODK Event Taxonomy

Event Type	Meaning
form start	The interview session began. Equivalent to SurveyCTO form-start.
question	The enumerator viewed a question field. This is the primary event for timing extraction.
group questions	A group of fields was displayed simultaneously (e.g., a repeat group header).
jump	The enumerator used the jump/go-to feature to navigate to a specific field.
end screen	The end-of-form confirmation screen was displayed.
save	The form was saved. Equivalent to SurveyCTO save event.
finalize	The form was finalized and submitted.

Table 4. ODK event taxonomy.

3.4 Example Log Rows

```
event,node,start,end
form start,,2024-03-26T10:00:00.000,2024-03-26T10:00:01.200
question,/data/consent,2024-03-26T10:00:01.200,2024-03-26T10:00:09.800
question,/data/household/q_age,2024-03-26T10:00:09.800,2024-03-26T10:00:19.600
question,/data/income/q_monthly,2024-03-26T10:00:19.600,2024-03-26T10:00:38.200
end screen,,2024-03-26T10:00:38.200,2024-03-26T10:00:39.000
```

3.5 Critical Differences from SurveyCTO

Important

ODK timestamps are ISO 8601 strings, not epoch integers. The ODKParser must convert these to milliseconds using datetime parsing before computing durations. Failure to do this will result in incorrect or null duration_ms values in the CIR.

- ODK does not record constraint events. The constraint_trigger_count feature (Q-09) cannot be computed from ODK logs and will be set to null for ODK-sourced interviews.

- ODK's 'question' event covers what SurveyCTO splits into multiple event types. This means some granular behavioral signals available in SurveyCTO are not recoverable from ODK logs.
- ODK timestamps have coarser granularity in some versions — second-level rather than millisecond-level. This affects the precision of time_to_first_interaction (Q-02) and idle_time_ms (Q-07).
- The old-value and new-value columns, when present, allow edit_count (Q-03) to be computed. When absent, edit_count defaults to null for ODK logs.

4. Platform Comparison

The following table summarises the key differences between SurveyCTO and ODK audit log formats relevant to the pipeline.

Attribute	SurveyCTO	ODK
Timestamp format	Epoch milliseconds (integer)	ISO 8601 string (must convert)
Timestamp precision	Millisecond	Millisecond (some versions: second only)
Event model	Granular event-type model	Question-centric model
Constraint events	Yes — constraint event type	No — not recorded
Edit tracking	Via repeated question events	Via old-value/new-value (version-dependent)
Form-start event	form-start	form start (with space, no hyphen)
Form-end event	end	end screen or finalize
GPS columns	Always present (may be null)	Present when location enabled
Back-navigation	Detectable via repeated question events	Detectable via repeated question events
CIR compatibility	Full — all 40 features computable	Partial — Q-09 and Q-03 may be null

Table 5. SurveyCTO vs ODK audit log format comparison.

5. Common Intermediate Representation (CIR)

The CIR is a standardised Pandas DataFrame produced by both parsers. All feature extraction and anomaly detection modules operate exclusively on the CIR — they are never exposed to raw SurveyCTO or ODK log formats. This separation ensures that new platform parsers can be added without modifying any downstream component.

5.1 CIR Schema

Column	Type	Nullable	Description
`interview_id`	String	**No**	Unique identifier for the interview session. Derived from the submission UUID or filename.
`field_name`	String	**Yes**	The field identifier extracted from the XPath node (final path component). Null for form-level events.
`event_type`	String	**No**	Normalised event type. One of: form_start, question, constraint, save, form_resume, form_end.
`start_ms`	Integer	**No**	Event start time in milliseconds since Unix epoch. Always an integer after normalisation.
`end_ms`	Integer	**Yes**	Event end time in milliseconds since Unix epoch. Null for instantaneous events.
`duration_ms`	Integer	**Yes**	Computed as end_ms minus start_ms. Null when end_ms is null or end_ms < start_ms.

Table 6. Common Intermediate Representation schema.

5.2 Normalised Event Types

Both parsers map their platform-specific event names to a common set of normalised event_type values used in the CIR:

CIR event_type	SurveyCTO source	ODK source
`form_start`	form-start	form start
`question`	question	question
`constraint`	constraint	(not available — null)

save	save	save
form_resume	form-resume	(not available)
form_end	end	end screen, finalize

Table 7. Normalised event type mapping.

5.3 Example CIR Output

The following shows a correctly formed CIR DataFrame for one interview:

interview_id	field_name	event_type	start_ms	end_ms	duration_ms
INT-0042	null	form_start	1711440000000	1711440001200	1200
INT-0042	consent	question	1711440001200	1711440009800	8600
INT-0042	q_age	question	1711440009800	1711440017400	7600
INT-0042	q_age	constraint	1711440014200	1711440014200	0
INT-0042	q_age	question	1711440014200	1711440019600	5400
INT-0042	q_monthly	question	1711440019600	1711440038200	18600
INT-0042	null	form_end	1711440038200	1711440039000	800

5.4 Minimum Log Requirements

For the pipeline to compute all features correctly, an audit log must satisfy the following minimum requirements:

- At least one form_start event and one form_end event must be present.
- At least five question events must be present (fewer makes statistical features unreliable).
- start_ms must be non-null and positive for all rows.
- end_ms must be greater than or equal to start_ms for question events (rows where end_ms < start_ms are dropped with a warning).
- The node/field_name column must be non-null for question events.

Warning

Interviews that do not meet the minimum log requirements are not dropped entirely. They are processed as far as possible, and features that cannot be computed are set to null. The interview receives a data_quality_warning flag in the output dataset. This behaviour is configurable.

6. Feature Computability by Platform

The following table documents which of the 40 features can be fully computed from SurveyCTO logs, partially computed, or are unavailable due to platform limitations.

ID	Feature	SurveyCTO	ODK	Notes
Q-01	response_time_ms	✓ Full	✓ Full	
Q-02	time_to_first_interaction	✓ Full	⚠ Partial	ODK second-level precision only
Q-03	edit_count	✓ Full	⚠ Partial	ODK requires old/new-value columns
Q-04	revisit_flag	✓ Full	✓ Full	
Q-05	revisit_count	✓ Full	✓ Full	
Q-06	field_completion_ratio	✓ Full	✓ Full	
Q-07	idle_time_ms	✓ Full	⚠ Partial	ODK second-level precision only
Q-08	keystroke_count	✓ Full	✗ N/A	ODK does not record keystrokes
Q-09	constraint_trigger_count	✓ Full	✗ N/A	ODK does not record constraints
Q-10	first_to_last_interaction_duration	✓ Full	⚠ Partial	Precision limited in some ODK versions
S-01 to S-10	All section-level features	✓ Full	✓ Full	Derived from question events
V-01 to V-10	All survey-level features	✓ Full	✓ Full	Derived from form_start/form_end
T-01 to T-10	All statistical features	✓ Full	✓ Full	Computed from Q-01 values

Table 8. Feature computability by platform.

7. Adding a New Platform Parser

The pipeline is designed so that adding support for a new platform requires only a new parser class. No changes to the feature extraction or anomaly detection modules are required, as these operate exclusively on the CIR.

To add a new platform parser:

1. Create a new class that inherits from `ParadataParser`.
2. Implement the `parse(filepath)` method to read the platform-specific CSV and return a CIR-compliant Pandas DataFrame.
3. Map all platform event types to the normalised CIR event_type values defined in Table 7.
4. Convert all timestamps to epoch milliseconds and store in `start_ms` and `end_ms`.
5. Compute `duration_ms` as `end_ms - start_ms`. Set to null where `end_ms` is null or negative.
6. Document any features that cannot be computed from the new platform's logs in this guide under a new section.

Design Principle

The CIR is the contract between the parser layer and the feature layer. As long as a parser produces a DataFrame that conforms to the CIR schema in Table 6, all 40 features and both anomaly detection layers will function correctly without any modification.

8. Glossary

Term	Definition
Audit log	A CSV file automatically generated by SurveyCTO or ODK that records every form interaction event during an interview, including timestamps and field identifiers.

CIR	Common Intermediate Representation. The standardised Pandas DataFrame format produced by all parsers and consumed by all feature extraction and detection modules.
Duration_ms	The computed duration of an event in milliseconds, calculated as end_ms minus start_ms.
Epoch milliseconds	A timestamp expressed as the number of milliseconds elapsed since 1 January 1970 00:00:00 UTC. Used natively by SurveyCTO; requires conversion from ISO 8601 for ODK.
Event taxonomy	The complete set of named event types recorded by a platform. Differs between SurveyCTO and ODK.
ISO 8601	An international standard for representing dates and times as strings (e.g., 2024-03-26T10:00:01.200). Used by ODK for timestamps.
Node	The XPath identifier of a form field (e.g., /data/household/q_age). Used by both SurveyCTO and ODK to identify which field an event relates to.
Paradata	Process data automatically captured by digital survey platforms during data collection, recording how interviews are conducted rather than the responses themselves.
Parser	A class that reads a platform-specific audit log CSV and produces a CIR-compliant DataFrame.
XPath	A notation for identifying elements in an XML document structure. Used by SurveyCTO and ODK to address form fields hierarchically.

Table 9. Glossary of terms used in this document.